BHARATCARES MACHINE LEARNING & APPLIED AI INTERNSHIP

PROJECT CONCEPT NOTE: LEARNMATE AI

Subtitle: Your Personal AI Learning and Student Success Companion
Theme: Machine Learning, Applied AI, and Agentic Multi-Agent Systems
Candidate: Joel Francis (B.Tech CSE) | **UN SDG Alignment:** Goal 4 — Quality Education | **Program Dates:** 24 August – 30 September 2026

1. Executive Summary

LearnMate AI is an intelligent, offline-capable education technology platform designed to bridge learning gaps for computer science students in underserved and resource-constrained environments. By combining empirical diagnostic testing, a trained Scikit-Learn Random Forest classification model (88.3% accuracy), an adaptive milestone study planner, and a 5-agent local orchestrator, LearnMate AI delivers personalized, pedagogical support without relying on expensive, cloud-hosted API subscriptions. The solution directly advances UN Sustainable Development Goal 4 (Quality Education) by providing equitable, transparent, and scalable academic guidance.

2. Background and Problem Statement

Modern university engineering curricula demand proficiency in programming, algorithms, and artificial intelligence. However, students frequently encounter three major hurdles: (1) An inability to identify precise conceptual gaps early; (2) Inconsistent study discipline and poor time-management habits; and (3) Inaccessible, costly 1-on-1 personalized tutoring. Commercial AI tutoring products rely heavily on paid closed-source LLMs that often produce hallucinations, lack persistent pedagogical tracking, and present significant recurring financial barriers for students from Tier-2 and Tier-3 institutions.

3. Target Users & SDG 4 Impact

The primary beneficiaries are undergraduate computer science and engineering students, particularly those in underserved colleges lacking personalized mentoring resources. LearnMate AI democratizes academic mentorship by operating completely free of proprietary API tokens, offering transparent machine learning predictions, and fostering self-regulated study consistency.

4. System Architecture

The system employs a modular, high-reliability architecture:
• **Frontend:** Responsive React 19 single-page interface with TypeScript, Vite, Tailwind CSS v4, Lucide icons, and Recharts visualization.
• **Backend API:** Asynchronous Python FastAPI microservice with Pydantic request validation and CORS middleware.
• **Data Persistence:** SQLite3 embedded relational database tracking user profiles, quiz attempts, study plans, and agent audit trails.
• **ML Inference:** Scikit-Learn supervised classification pipeline with Joblib artifact serialization.
• **Agent Collective:** Multi-agent orchestration engine with deterministic execution planning and transparent fallback recovery.

5. Machine Learning Methodology

A Scikit-Learn **RandomForestClassifier** (100 estimators, max depth 6) was trained to classify students into three pedagogical support categories: (0) *Needs foundational revision*, (1) *Developing understanding*, and (2) *Ready for advanced practice*.

- **Feature Inputs:** (1) Previous Quiz Score, (2) Recent Quiz Average, (3) Practice Task Completion Rate, (4) Total Quiz Attempts, (5) Cumulative Study Time Hours, and (6) Performance Trajectory Trend.
- **Model Evaluation:** Evaluated on an 80/20 stratified split (960 train / 240 test). The primary Random Forest achieved **88.33% Accuracy** and **0.8809 Macro F1-Score**, substantially exceeding the Decision Tree baseline (86.67%). Feature importance analysis revealed Recent Quiz Average (32.4%) and Practice Completion Rate (24.1%) as primary drivers.
- **Transparent Provenance Disclosure:** Training was performed on a synthetic educational dataset of 1,200 trajectories calibrated with realistic noise to evaluate student pacing. The model output is strictly utilized to tailor study pacing, never to assess general human intelligence.

6. Multi-Agent Agentic Workflow

LearnMate AI implements a multi-agent collective coordinated by a Master Orchestrator:
1. **Tutor Agent:** Retrieves curated concept explanations and code patterns using local TF-IDF matching across verified knowledge cards.
2. **Quiz Agent:** Dynamically retrieves domain-specific diagnostic questions, checks submissions, and generates instant explanatory feedback.
3. **Learning Analyst Agent:** Analyzes quiz attempt logs, calculates mastery velocity, flags weak subjects, and invokes the ML classifier.
4. **Study Planner Agent:** Synthesizes diagnosed gaps, student deadlines, and daily hours into a structured 5-day milestone checklist.
5. **Career Guide Agent:** Correlates verified competencies with industry roles (AI/ML Engineer, Data Scientist, Full-Stack AI) and recommends portfolio capstone projects.

7. Ethical Considerations & Limitations

• **Privacy by Design:** Zero personal identifying data or sensitive information is collected or transmitted externally.
• **Hallucination Prevention:** When student queries fall outside curated topics, the Tutor Agent provides an honest fallback rather than fabricating sources.
• **Synthetic Data Notice:** All evaluation metrics reflect the synthetic prototype dataset and will undergo longitudinal validation in live classrooms during future enhancements.

Submission Status: *Completed, tested, and verified for BharatCares Internship Evaluation.*